COMPLETE DOCUMENTATION AND SOURCE CODE

DPI Studio

Print-ready images at 300-600 DPI - from one private, offline HTML file. No installation. No account. No uploads.

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Document: Full documentation - overview, applications, workflow, technical method, features, benefits, limitations, uniqueness, deployment and operating-system guides, glossary and troubleshooting.

Appendix A: The complete source code of DPI Studio (139 lines), reproduced verbatim with line numbers, so the app can be rebuilt from this document alone.

What the tool honestly does: DPI Studio writes correct print-density (DPI) metadata into PNG and JPEG files and, when asked, resizes images by interpolation to match a chosen print width. It does not enhance image quality, add detail, or fix blur - and this document says so wherever it matters.

CONTENTS

What this document covers

1. Overview - what DPI Studio is 3

2. Applications - where it is used 4

3. How it works - DPI, pixels and print size 5

4. Workflow - step by step, with flow diagram 6

5. Technical method - how the code does it 7

6. Features - full reference table 9

7. Benefits 10

8. Limitations - stated plainly 11

9. Uniqueness - how it differs 12

10. Deployment on other machines 13

11. Operating systems - opening the app anywhere 14

Appendix A - Complete source code (139 lines, verbatim) 15

Appendix B - Glossary and troubleshooting FAQ 23

About this document

This document is the complete record of DPI Studio: what it does, how it works, where it helps, where it stops, and the full code that makes it run - for the everyday user who wants a print-ready image, and for the technical reader who wants to verify, byte by byte, what the file does.

Nothing in DPI Studio is a black box. The app is a single HTML file. Its entire logic - the PNG and JPEG writers, the ZIP builder, the verification step - is printed in Appendix A. If the two were separated, the app could be rebuilt from Appendix A alone: copy the code into a text file, save it as `DPI-Studio.html`, and open it in a browser.

> **Creator credit.** DPI Studio was designed and generated by **Dr. Asif Ahmad**. This documentation is credited to him throughout, on the cover and in every page footer.

Conventions used here

- **DPI** means dots per inch - the print density stored in the image file. **PPI** (pixels per inch) is the same idea described from the pixel side; Appendix B explains both.
- Measurements are given in inches first, with centimetres alongside, matching the app itself.

SECTION 1

Overview - what DPI Studio is

DPI Studio is a private, browser-based tool that creates print-ready copies of PNG and JPEG images at a print density the user chooses - any whole number from 300 to 600 DPI.

It exists for a small, common, frustrating job. A journal, a printer, a university office or a conference organiser asks for an image "at 300 DPI", and the image on your computer either carries no density information or carries the screen default of 72 or 96 DPI. The picture may be perfectly good - it simply does not *say* the right number, and may not have the pixel dimensions the print size needs. DPI Studio fixes precisely that, and nothing more.

One file, nothing else

The whole application is a **single offline HTML file** (about 33 KB). There is:

- **No installation** - double-click the file and it opens in a web browser.
- **No account and no sign-in** - there is nothing to register for.
- **No uploads** - images are processed inside the browser, on the device, and never leave it. The interface states this plainly: "Private by design. Images are processed in your browser and are never uploaded."
- **No server and no dependencies** - no runtime, library or internet connection is required once the file is on the machine.

What it produces

For every image processed, DPI Studio produces a **fresh copy** - the original file is never modified ("Your original files stay untouched"). The copy carries:

- the chosen print density (300-600 DPI) written into the file's own metadata, in the correct binary location for its format;
- optionally, new pixel dimensions calculated from a print width the user chooses (see Section 3);
- the user's choice of output format - PNG (lossless, keeps transparency) or JPEG (smaller file, white background);
- a built-in **readback verification**: before any download begins, the app re-reads the finished file and confirms the DPI metadata is really there and correct. If verification fails, no file is downloaded and an error is shown instead.

Who it is for

DPI Studio was built for people who are not image specialists - researchers, students, teachers, office staff and small publishers - who need a safe, honest way to comply when told "send it at 300 DPI". Controls are labelled in plain language, the output size is previewed live, and the one thing the tool cannot do - invent detail that is not in the picture - is explained on the app's own screen, not hidden.

> **In one sentence:** DPI Studio takes an image you already have, stamps it with the correct print density, resizes it by interpolation only when you ask it to, verifies the result, and hands you the copy - all inside your browser, with your files never leaving your device.

SECTION 2

Applications - where DPI Studio is used

Anywhere a person or an organisation asks for an image "at 300 DPI or higher", DPI Studio is the quick, private way to prepare it.

Academic publishing

- **Journal figures.** Most journals require figures at 300 DPI or more for photographs and higher for line art. A researcher can take each figure, set 300, 450 or 600 DPI, resize it to the journal's column width, and download a compliant copy in minutes.
- **Conference papers and posters.** Conference templates and poster printers specify density and physical size together - exactly the combination DPI Studio calculates live (pixels, inches and centimetres on one screen).
- **Thesis printing.** Universities commonly require embedded images at print density for the bound thesis. Batch mode prepares a whole chapter's figures at once and delivers them as a single ZIP.

Print and publishing work

- **Poster and banner printing.** Choose the real print width (for example 24 inches) and the printer's DPI, and the app works out the pixel dimensions the file needs.
- **Press and print publishing.** Newsletters, brochures, book covers and institutional reports prepared for a commercial printer, where the pre-press check asks for density metadata, not just pixels.
- **Photo printing.** Home and lab photo prints come out at the intended size when the file's density matches the print size instead of the screen default.

Organisation and teaching

- **Archiving print-ready copies.** Keep a folder of verified 300 DPI masters alongside originals, so a future reprint never starts from a screen-density file.
- **Batch-converting image sets.** Up to 10 images at a time - a product set, a set of certificate scans, an event's selected photographs - converted with one setting and downloaded as one ZIP.
- **Teaching print-resolution concepts.** The live output summary makes the DPI-pixels-print-size relationship visible: move the slider and the print size changes; enable resizing and the pixel count changes. It is a working demonstration of the idea, not just a converter.
- **Office and administrative submissions.** Any portal, tender or institutional form that specifies a density for uploaded images, photographs or scanned signatures.

> **Fit, honestly stated:** DPI Studio prepares density and size. It is not the tool for colour-managed commercial pre-press (it works in RGB and does not handle CMYK - see Section 8), and it cannot rescue a small, blurry original. Section 3 explains why, in plain terms.

SECTION 3

How it works - DPI, pixels and print size

Three ideas explain everything DPI Studio does. Once they are clear, every control in the app makes sense.

Idea 1: Pixels are the picture; DPI is an instruction

An image file holds a grid of pixels - say 3,000 wide by 2,000 tall. That grid *is* the picture; all the detail lives there. Separately, the file can carry metadata: its **print density**, in DPI (dots per inch). DPI is not part of the picture. It is an instruction to printing software: "place this many pixels into each inch of paper."

Idea 2: Print size is pixels divided by DPI

```
print width (inches) = pixel width ÷ DPI
```

The same 3,000-pixel-wide image prints at three sizes depending only on its DPI instruction:

Pixel width	DPI instruction	Printed width	Typical use
3,000 px	300 DPI	10 inches (25.4 cm)	Full-page journal figure
3,000 px	450 DPI	6.67 inches (16.9 cm)	Detailed figure, column-plus width
3,000 px	600 DPI	5 inches (12.7 cm)	Small, dense print - line art

This is why simply "raising the DPI" makes the same image print *smaller*, not better: the pixels are squeezed into fewer inches. The picture itself has not changed.

Idea 3: Raising DPI does not add detail or fix blur

DPI Studio states this on its own screen: changing the DPI number relabels how densely pixels are placed. It does not create new pixels, recover detail the camera never captured, or sharpen a blurred image. A soft original yields the same soft image carrying a correct, honest density value.

The resize option: new pixels for a chosen print size

Sometimes a publisher wants *both* a print size and a density - "6 inches wide at 300 DPI". That combination decides the pixel count:

```
new pixel width = print width (inches) × DPI
```

So 6 inches at 300 DPI needs 1,800 pixels across; at 600 DPI, 3,600. With "Resize for a chosen print width" (inches or centimetres), the app calculates the new width, scales height in the same proportion so the picture is never stretched, and redraws the image using the browser's high-quality smoothing. Enlarging this way is **interpolation**: new pixels are estimated from neighbours. The file measures correctly and prints at the requested size and density - but the added detail is estimated, never recovered. The app says so at the moment it matters: "Enlargement: 1.50x. New pixels are interpolated; missing detail cannot be restored."

> **The honest summary:** keep-dimensions mode changes only the instruction; resize mode changes the instruction *and* the pixel grid, by interpolation. Neither enhances image quality.

SECTION 4

Workflow - from image to verified download

One pass through the app takes under a minute per batch. The flow below is the exact path the interface leads the user through.

1. **Add images**
   Drag and drop onto the drop zone, or click to browse. Up to 10 images: JPG, PNG or WebP, 25 MB each, 100 MB per batch. Animated PNG/WebP files are refused - still images only.

↓

2. **Preview and inspect the original**
   The selected image is previewed, with a thumbnail queue for the batch. Three facts are shown: pixel size, **detected DPI** (read from the file's metadata, or "Not detected"), and file size.

↓

3. **Choose the target DPI - 300 to 600**
   Drag the slider, tap a preset (300 Standard, 450 Detailed, 600 High density), or type an exact whole number. A short description under the number explains the choice in plain words.

↓

4. **Optionally resize for a print width**
   Either keep the original pixel dimensions (the default), or switch to "Resize for a chosen print width" and enter the width in inches or centimetres. Height follows automatically, keeping the original proportions.

↓

5. **Choose the output format**
   PNG for a lossless copy that preserves transparency, or JPG for a smaller file in which transparent areas are placed on white.

↓

6. **Check the live output summary**
   Before anything is downloaded, the summary shows the output pixel dimensions and the resulting print size in inches and centimetres - plus an enlargement warning whenever resizing will interpolate new pixels.

↓

7. **Download - single image or batch ZIP**
   Download the selected image on its own, or, with two or more images loaded, download them all as one ZIP file named with the chosen DPI.

↓

8. **Readback verification (automatic, behind the scenes)**
   Every export is re-read before release: the app parses the finished file, reads its DPI back, and compares it with the requested value. Only a passing file is downloaded; a failure shows an error instead.

SECTION 5 - TECHNICAL METHOD

How the code actually does it

DPI Studio does not relabel files or trust the browser to set density. It edits the binary structure of each format directly, then proves the edit worked. This section describes the real mechanisms, exactly as implemented in Appendix A.

The core engine: DpiCore

All file-format logic lives in one script block exposing a small library, `DpiCore`: `crc32`, `pngDpi`, `jpegDpi`, `readDpi`, `dimensions`, `zip`, `pngChunks` and the `validDpi` guard. The interface code never touches file bytes itself - it asks DpiCore. That separation is why the engine can also run outside the browser (it exports itself for module systems when present).

Writing DPI into a PNG: the pHYs chunk

A PNG file is a signature followed by chunks, each with a length, a four-letter type, its data, and a CRC32 checksum. Print density lives in the **pHYs** chunk, storing pixels per unit - and PNG's unit is the *metre*, not the inch. The conversion:

```
pixels per metre = DPI ÷ 0.0254  (rounded to a whole number)
```

- At 300 DPI that is 11,811 pixels per metre; at 600 DPI, 23,622.
- The writer builds a 9-byte pHYs payload: horizontal value, vertical value (identical, so prints are never distorted), and a unit byte of 1, "per metre".
- It computes the **CRC32 checksum** over the chunk type plus data, using a table-driven CRC32 implementation identical in construction to the one the ZIP writer uses.
- It then **splices** the file back together: the PNG signature, every original chunk *except* any existing pHYs chunk (dropped, so a file can never carry two conflicting densities), and the new pHYs chunk inserted **immediately after the IHDR header chunk**, where the specification expects it.
- An invalid PNG, a missing end marker or a missing header produces a plain-language error, never a corrupt file.

Writing DPI into a JPEG: the JFIF APP0 segment

A JPEG opens with a marker (FF D8) followed by segments. Density lives in the **JFIF APP0** segment: a units byte plus X and Y density values. The code first walks the segment structure, validating every marker and length. Then:

- **If a JFIF APP0 segment exists**, it is edited in place: units byte set to **1 (dots per inch)**, and the chosen DPI written as a 16-bit value into both X- and Y-density fields.
- **If no JFIF segment exists** (some cameras and editors omit it), the writer builds a complete new 16-byte APP0 segment - "JFIF" identifier, version 1.2, units 1, the chosen DPI in both density fields - and inserts it immediately after the opening marker.
- The compressed image data (the scan segment) is carried through untouched; only header bytes change in keep-dimensions mode.

> **Why this matters:** many simple "DPI changers" only rename a file. DPI Studio edits the actual standard fields printers and pre-flight checks read - which is why its output passes verification.

Verification, resizing, ZIP and limits

Readback verification - trust, then check

Every export passes through `readDpi` before download. The reader parses the finished bytes from scratch: for PNG it finds the pHYs chunk and converts pixels-per-metre back to DPI (x 0.0254); for JPEG it reads the JFIF density, honouring the units byte (per inch, or per centimetre, converted by 2.54). The export is released only if the value read back matches the request within 0.02; otherwise it aborts with "The DPI metadata could not be verified." The same reader shows each original's "Detected DPI" on load.

Canvas decode and resize

To change formats or pixel dimensions, the image is decoded into a browser canvas. Resizing draws it at the calculated size with `imageSmoothingEnabled` on and `imageSmoothingQuality` set to `'high'` - the best interpolation the browser offers. For JPEG the canvas is first filled white (JPEG has no transparency). The canvas is encoded (quality 0.98 for JPEG) and the bytes pass through the density writer above - so even a format conversion ends with verified metadata. The planner (`dimensions`) enforces the envelope: no side over 16,384 px, no image over 40 MP.

The hand-rolled ZIP writer

Batch downloads use no library. The built-in ZIP writer emits **stored (uncompressed)** entries: a local header (signature 0x04034b50) and the raw bytes per file, then a central directory and end record (0x02014b50, 0x06054b50), with a CRC32 per file, DOS timestamps and UTF-8 filename flags. Stored entries keep the writer small; PNG/JPEG data barely compresses anyway, so the ZIP is a container, not a compressor.

The limits, as coded

Limit	Value in the code	Reason
DPI range	Whole numbers, 300-600 only	The tool's stated purpose; anything else is rejected
Images per batch	10	Keeps browser memory and the ZIP manageable
Size per image	25 MB input; 40 megapixels decoded	Browser canvas and memory safety
Input batch total	100 MB	Same - one batch cannot exhaust the tab
ZIP output cap	200 MB	Checked while preparing; otherwise download individually
Output ceiling	16,384 px per side; 40 MP total	Enforced by the dimension planner before any resize
Animated images	Animated PNG (acTL) and animated WebP (VP8X flag) rejected	Stamping an animation the tool cannot faithfully rewrite would corrupt it
Accepted inputs	JPG, PNG, WebP (still)	WebP is decoded as input; output is always PNG or JPG

SECTION 6

Features - the full reference

Every capability in the shipped app, in one table. If a capability is not listed here, DPI Studio does not claim it.

Feature	What it does
300-600 DPI selection	Any whole-number density in range, set by slider, preset (300 / 450 / 600), or typing the exact number. Out-of-range values are rejected with a plain message.
True binary metadata writing	Writes the PNG pHYs chunk or the JPEG JFIF density fields directly in the file bytes - the values real printers and pre-flight checks read.
Readback verification	Every export is re-parsed and its DPI read back and compared before download. A mismatch aborts the download.
Detected-DPI display	Shows the density already stored in each original image, so the user can see the "before" (including "Not detected").
Keep dimensions or resize	Relabel density only, or calculate new pixel dimensions from a target print width in inches or centimetres, with proportions preserved automatically.
Live output summary	Output pixel dimensions and print size (inches and centimetres) update as settings change - before anything is downloaded.
Enlargement honesty note	When a resize enlarges beyond about 1.01x, the summary states the factor and that new pixels are interpolated and detail cannot be restored.
PNG and JPEG output	PNG is lossless and preserves transparency; JPEG is smaller, encoded at high quality (0.98), with transparency placed on white.
Batch processing with ZIP	Up to 10 images processed with one set of settings and downloaded as a single ZIP, built by the app's own dependency-free ZIP writer.
Drag-and-drop and browse	Drop files anywhere in the window or browse; a thumbnail queue switches between loaded images; any image can be removed individually.
Originals never modified	All work happens on in-memory copies. The files on disk are untouched.
Privacy by design	No uploads, no account, no analytics, no network calls. Processing is entirely in the browser.
Offline single file	The complete app is one HTML file (about 33 KB) that runs from a local disk, USB stick or phone with no internet.
Guard rails	Specific errors for oversize files, animated images, invalid formats, excessive output size and ZIP overflow - each saying what to do instead.
Accessible touches	Keyboard-focus outlines, labelled controls, a live-region toast for status, and a layout that adapts to phone-width screens.

SECTION 7

Benefits - what the user actually gains

The benefits of DPI Studio follow directly from three design decisions: it runs locally, it costs nothing, and it tells the truth about what it is doing.

Privacy that is structural, not promised

Online converters ask the user to upload images to a stranger's server and trust a privacy policy. DPI Studio makes that trust unnecessary: the processing happens in the browser, the images never leave the device, and there is no server to send them to. Unpublished research figures, student photographs, identity documents and commercially sensitive artwork can be prepared without exposure. This matters most for exactly the academic and institutional users the tool serves.

No cost, no installation, no friction

- **Free, in full.** Every feature - batch mode, ZIP export, the whole 300-600 range - is in the single file. There is no trial, tier, watermark or per-file fee.
- **Nothing to install or update.** The app is the file. It works the same on the day it is copied as years later, and it cannot be broken by an expired subscription or a changed web service.
- **Works offline.** On a plane, in a field station, in a classroom with no connectivity - anywhere a browser opens, the tool runs.
- **Works on the machine in front of you.** Because it needs only a browser, the same file serves a Windows office PC, a Mac, a Linux workstation, a Chromebook and a phone (Section 11).

Confidence in the output

- **Verified, not assumed.** The readback check means a downloaded file is a *proven* 300 DPI file, not a hoped-for one. Users can submit without a second tool to double-check.
- **The right size, seen in advance.** The live summary prevents the two classic mistakes - a file that prints far smaller than expected, and an accidental enormous enlargement - before any download happens.
- **Originals are safe.** Experiment freely; the source files are never overwritten.
- **Batch consistency.** A whole figure set leaves with identical settings in one ZIP, instead of ten files converted ten slightly different ways.

Beginner-friendly by design

The interface is two numbered steps - "Your images" and "Choose your DPI" - with presets named Standard, Detailed and High density, plain-language explanations on screen (including a permanent panel titled "What does changing DPI do?"), and error messages that say what went wrong and what to do. A user who has never heard of a pHYs chunk can still finish the job correctly, while the expert can audit exactly how it was done in Section 5 and Appendix A.

> **The benefit underneath the others:** DPI Studio removes a small recurring blocker from academic and print work - permanently, privately and for free - without pretending to be a larger tool than it is.

SECTION 8

Limitations - stated plainly

These limits are part of the design record. A user who reads this page will never be surprised by the tool.

It cannot add detail or fix blur

This is the fundamental limit, and it is physics, not a missing feature. The pixels in the original are all the information there is. Raising the DPI number only changes how densely those pixels are placed on paper. Resizing upward interpolates - it estimates new pixels from neighbouring ones - which makes a file that *measures* correctly but contains no more real detail than it started with. A small or out-of-focus original will produce a correctly-labelled small or out-of-focus copy. No setting in DPI Studio changes that, and the app says so on screen and in its output summary.

Metadata and colour

- **EXIF metadata is not carried over.** Because output is produced by decoding and re-encoding the image, camera data, GPS tags, timestamps and other EXIF fields in the original are not copied into the new file. The interface states this under the format selector. Keep the original when that data matters.
- **No CMYK or press colour management.** Output is standard RGB PNG or JPEG. Work destined for a commercial press that requires CMYK separations or specific ICC profiles needs a pre-press tool; DPI Studio does not convert colour spaces.
- **Format conversion re-encodes.** A JPEG saved as JPEG is decoded and re-encoded (at high quality, 0.98); PNG output is lossless relative to the decoded image.

Practical ceilings

- **Browser memory is the ceiling.** Very large images are refused by design: 25 MB per input file, 40 megapixels decoded, 16,384 pixels on any output side, 100 MB per input batch, 200 MB per ZIP. These are generous for figures and photographs, but gigapixel scans and huge print-shop files belong in desktop software.
- **Animated images are unsupported.** Animated PNG and animated WebP files are detected and rejected rather than risk corrupting them.
- **Input formats are JPG, PNG and WebP; output is PNG or JPG only.** TIFF - common in publishing - is not handled in either direction.
- **It is not a photo editor.** No cropping, retouching, colour correction, sharpening or compression tuning. It does one job: density, size and format for print.
- **Verification covers DPI, not everything.** The readback check proves the density metadata. It does not judge whether the picture is good, in focus or the right picture.

> **Used within these limits, the tool is exact.** Used outside them - as a quality enhancer, a CMYK pre-press stage or an editor - it will disappoint, and this page exists so that nobody has to discover that by trial and error.

SECTION 9

Uniqueness - how DPI Studio differs

DPI converters are common. What is uncommon is the combination DPI Studio ships in: real binary editing, self-verification, total privacy and total portability, in 33 KB, for free.

DPI Studio	Typical online DPI converter	Typical desktop software
One HTML file; runs offline forever	A website; needs internet and the service staying online	A large installation, often paid, updated on the vendor's schedule
Images never leave the device - there is no server	Images are uploaded to the operator's server	Local, but inside a heavyweight application
Writes true format-level metadata (PNG pHYs chunk; JPEG JFIF fields)	Varies; some only change a display value or re-encode loosely	Yes - buried inside a professional workflow
Readback verification on every single export	Essentially unheard of - the file is returned unchecked	No equivalent automatic proof step
Free, complete, no account	Often freemium - limits, queues, watermarks, sign-up prompts	Licence or subscription for full features
States its core limit on its own screen	Marketing often implies quality improvement	Manuals document it; the interface does not volunteer it
Batch ZIP with a hand-rolled, dependency-free writer	Batch is commonly a paid tier	Available, via export presets

The five distinctives

1. **Single-file offline app.** The entire product - interface, format writers, ZIP engine - is one portable HTML file. It can be emailed, carried on a USB stick or archived next to this document, and it will still run years from now.
2. **True binary-level metadata writing.** The tool edits the PNG chunk structure and JPEG segment structure directly, with correct CRC32 checksums and unit conversions - not a relabelled export that some printers silently ignore.
3. **Built-in readback verification.** DPI Studio checks its own work on every file and refuses to hand over an export whose density does not read back correctly. The converter is also the checker.
4. **Privacy by design, not by policy.** There is no upload because there is nowhere to upload to. The guarantee is architectural.
5. **Honesty as a feature.** The enlargement warning, the on-screen explanation and this limitations section are part of the product. For academic users - whose publishers will inspect the result - a tool that does exactly what it claims is the professional choice.

SECTION 10

Deployment - putting DPI Studio on other machines

Deployment means copying one file. There is no server, runtime, database or dependency to install - on any machine, ever.

Method 1: Copy the file (the standard way)

1. Copy `DPI-Studio-Offline.html` by USB stick, email attachment, shared drive or any cloud folder (Google Drive, OneDrive, Dropbox).
2. Save it anywhere convenient - Desktop or Documents are ideal.
3. Double-click it. It opens in the default browser, fully functional and fully offline.

Method 2: Desktop shortcut

- **Windows:** right-click the file → Send to → Desktop (create shortcut). Or in Chrome/Edge, open the file, then use the browser menu → Save and share / More tools → Create shortcut, to give it its own app-like icon and window.
- **macOS:** drag the file to the Dock's right side, or in Chrome use File → Save As / the "Install" style shortcut options to pin it like an app.

Method 3: Shared network or intranet

Place the file on a departmental shared drive or internal web server. Every user opens the same copy in their browser; each user's images are still processed only on their own machine, because the file runs entirely client-side. Updating the whole department is replacing one file.

Method 4: Static web hosting

Upload the single file to any static host (a university web folder, GitHub Pages, Netlify, a plain web server). No server-side code runs and nothing is collected; visitors' images are processed in their own browsers. Rename it to `index.html` if it should be the site's front page.

Method 5: Phone or tablet

Send the file to the device (email, cloud drive, cable), save it to local storage, and open it with the browser's "Open in Chrome / Safari" option or a file manager. Add it to the home screen for one-tap access. All processing stays on the device.

> **Version note:** the app has no update mechanism, by design - it cannot phone home. To "upgrade" a deployment, replace the file with the newer copy. The complete source in Appendix A is the permanent record of this version.

SECTION 11

Operating systems - the same file, everywhere

DPI Studio runs wherever a modern browser runs. The file never changes - only the way it is opened differs slightly by system.

System	How to open	Notes
Windows 10 / 11	Double-click the file (opens in the default browser, usually Edge). Or right-click → Open with → Chrome / Edge / Firefox.	Works fully offline. Any modern browser is fine; Chrome or Edge give the smoothest large-file handling.
macOS	Double-click (opens in Safari), or right-click → Open With → Chrome / Safari / Firefox.	Downloads land in the Downloads folder as usual. Safari is fully supported.
Linux	Double-click in the file manager, or open the browser and press Ctrl+O and pick the file.	Firefox and Chromium both run it unchanged. No packages, no terminal, no permissions needed.
ChromeOS	Save the file in the Files app (Downloads or Google Drive), double-tap / double-click to open in Chrome.	Ideal fit: the whole system is browser-centred, and the app works with the connection off.
Android	Save the file to the device, then open it from the Files / My Files app and choose Chrome; or open Chrome → menu → Downloads and tap the file.	Use "Add to Home screen" from Chrome's menu for app-like access. Batch sizes are smaller in practice - phone memory is the limit.
iOS (iPhone / iPad)	Save the file in the Files app, tap it, and open it in Safari; or store it in a cloud drive app and use "Open in Safari".	Works in Safari's browser engine. Keeping the file in iCloud Drive makes it available on all the user's Apple devices at once.

Browser notes

- **Any current browser works:** Chrome, Edge, Firefox and Safari all implement everything the app uses (canvas, file handling, downloads). No extensions or plug-ins are involved.
- **Offline is real offline:** once the file is on the device, turn the connection off and test - every feature, including batch ZIP, still works, because nothing is fetched.
- **The "Use offline" link hides itself:** when the app is already running from a local file, the header's offline-download link disappears, since the user already has the offline copy open.
- **Very old browsers** (pre-2020) may lack one of the modern features the code uses and show errors; any browser kept even loosely up to date is fine.

> **One file, six systems, zero installation:** that portability is why DPI Studio can be handed to a whole class, department or editorial office in a single email.

APPENDIX A

Complete source code - DPI Studio (offline)

The complete, unedited source of **DPI-Studio-Offline.html** - all 139 lines, reproduced verbatim with the original line numbers. To rebuild the app: copy this code into a plain text file, save it as `DPI-Studio.html`, and open it in any browser. Line 1 of the original file is a single very long line (9,960 characters: the document head and all CSS, minified onto one line in the original). It is shown wrapped, and split once across a page boundary, marked "1 cont." - every character is present, in order. Long lines elsewhere wrap the same way; the line number always refers to the original file.

File: DPI-Studio-Offline.html - 139 lines - listing begins

```
1 <!doctype html><html lang="en"><head><meta charset="utf-8"><meta name="viewport" content="width=device
-width,initial-scale=1"><meta name="description" content="Set image print resolution from 300 to 600 D
PI. A private, browser-based PNG and JPEG tool."><title>DPI Studio — Print-ready images</title><style
>:root{--bg:#f7f6f1;--paper:#fff;--ink:#283e38;--muted:#89948e;--line:#e7ebe4;--green:#486b5b;--warm:#
b19066}*{box-sizing:border-box}body{margin:0;background:var(--bg);color:var(--ink);font-family:Inte
r,'Segoe UI',Arial,sans-serif;font-size:14px}button,input,select{font:inherit}button,a{-webkit-tap-hig
hlight-color:transparent}a{color:inherit;text-decoration:none}button{cursor:pointer}button:disabled{cu
rsor:not-allowed;opacity:.45}[hidden]{display:none!important}header{max-width:1160px;margin:auto;paddi
ng:24px 32px;display:flex;align-items:center;justify-content:space-between;border-bottom:1px solid #e5
e8df}.brand{display:flex;align-items:center;gap:10px;font-size:21px;font-weight:750;letter-spacing:-.8
px}.brand-light{font-weight:400;color:#819086;margin-left:3px}.mark{width:34px;height:34px;border-radi
us:9px;background:#e6ece2;display:grid;place-items:center;color:#648270}.mark svg{width:23px;height:23
px}.header-links{display:flex;align-items:center;gap:28px}.private{font-size:12px;color:#8b9890}.priva
te i{width:6px;height:6px;display:inline-block;border-radius:50%;background:#75a183;margin-right:7px}.
offline-link{font-size:12px;font-weight:600;color:#607c6a}main{max-width:1100px;margin:auto;padding:0
30px}.intro{text-align:center;padding:43px 0 34px}.eyebrow{font-size:11px;letter-spacing:2px;font-weig
ht:650;color:var(--warm);margin:0 0 16px}h1{font-size:43px;line-height:1.13;letter-spacing:-2px;font-w
eight:650;margin:0 0 16px}h1 span{color:#85978b}.intro>p:last-child{font-size:12px;color:var(--muted);
margin:0}.workspace{display:grid;grid-template-columns:1.14fr 1fr;gap:22px}.image-panel,.settings-pane
l{background:var(--paper);border:1px solid var(--line);border-radius:18px;padding:26px}.panel-top{disp
lay:flex;justify-content:space-between;align-items:center;margin-bottom:25px;gap:12px}.panel-top>div{d
isplay:flex;align-items:center;gap:9px}.step{font-size:11px;font-weight:700;color:#93a391;width:23px;h
eight:23px;display:grid;place-items:center;border:1px solid #e3e9df;border-radius:50%}h2{font-size:14p
x;font-weight:600;letter-spacing:-.3px;margin:0}.quiet{font-size:12px;color:#7e9384;border:0;backgroun
d:transparent}.print-badge{font-size:7px;letter-spacing:1.1px;color:#a2aa9f}.dropzone{width:100%;min-h
eight:345px;border:1px dashed #cdd9c9;border-radius:12px;background:#f9fbf6;padding:35px 12px;display:
flex;align-items:center;justify-content:center;flex-direction:column;transition:.2s}.dropzone:hover,.d
ropzone.dragging{border-color:var(--green);background:#eef5e9}.upload-art{color:#93a78d;width:86px;hei
ght:86px;background:#edf2e5;border-radius:50%;display:grid;place-items:center;margin-bottom:20px}.uplo
ad-art svg{width:67px;height:67px}h3{font-size:17px;font-weight:550;letter-spacing:-.4px;margin:0}.dro
pzone>p{font-size:12px;color:#a1aaa1;margin:9px 0 23px}.dropzone strong{color:#6d8769;font-weight:55
0}.file-hint{font-size:11px;color:#a5ada1}.privacy-note{display:flex;align-items:center;gap:9px;color:
#a5b09f;margin-top:24px}.privacy-note svg{width:18px;height:18px;flex-shrink:0}.privacy-note p{font-si
ze:11px;line-height:1.6;color:#9da89b;margin:0;max-width:340px}.dpi-display>div{display:flex;align-ite
ms:baseline;gap:6px}.dpi-display input{padding:0;border:0;background:transparent;width:113px;font-siz
e:53px;line-height:1;letter-spacing:-2px;font-weight:650;color:var(--ink);outline:none;-moz-appearanc
e:textfield}.dpi-display input::-webkit-outer-spin-button,.dpi-display input::-webkit-inner-spin-butto
n{-webkit-appearance:none;margin:0}.dpi-display span{font-size:12px;color:#9ba89f;letter-spacing:1px}.
dpi-display p{font-size:12px;color:#96a294;margin:5px 0 21px}input[type=range]{width:100%;height:5px;a
ppearance:none;-webkit-appearance:none;border-radius:20px;background:linear-gradient(to right,var(--gr
een) 0%,#e6eadd 0%);outline:none;cursor:pointer;display:block}
```

Appendix A - Source code (continued) - lines 1 cont. to 3 of 139

```
input[type=range]::-webkit-slider-thumb{appearance:none;-webkit-appearance:none;width:15px;height:15px;border:3px solid #fff;background:var(--green);border-radius:50%;box-shadow:0 0 0 1px #bdcbb8,0 2px 5px #38514430}input[type=range]::-moz-range-thumb{width:10px;height:10px;border:3px solid #fff;background:var(--green);border-radius:50%;box-shadow:0 0 0 1px #bdcbb8}.range-labels{display:flex;justify-content:space-between;color:#a5afa2;font-size:12px;margin-top:10px}.presets{display:grid;grid-template-columns:repeat(3,1fr);gap:8px;margin:18px 0 25px}.preset{padding:11px 5px;background:white;border:1px solid #e6ebe1;border-radius:8px;color:#80907d;font-size:12px}.preset small{display:block;margin-top:4px;font-size:12px;color:#a4afa0}.preset.active{background:#f0f5e9;border-color:#b8cbb0;color:#4e714c}.preset.active small{color:#8aa080}.field-section{margin:19px 0}.field-label{font-size:12px;font-weight:600;color:#687d6b;display:block;margin-bottom:8px}select,input[type=number]:not(#dpi-number){width:100%;padding:11px 12px;background:#fbfcf8;border:1px solid #e2e8db;border-radius:8px;color:#6d7e6b;font-size:12px;outline:none}select:focus,input:focus-visible,button:focus-visible{outline:2px solid #93ad81;outline-offset:3px}.field-help{font-size:11px;color:#a0aa99;line-height:1.7;margin:7px 0 0}.output-summary{padding:14px 16px;border:1px solid #e8ecdf;border-radius:9px;background:#f8faf3;margin-top:22px}.output-summary>div{display:flex;align-items:center;justify-content:space-between;gap:12px;margin-bottom:9px}.output-summary>div:last-of-type{margin-bottom:0}.output-summary span{font-size:11px;color:#a0ac97}.output-summary strong{font-size:12px;font-weight:500;color:#72836d;text-align:right}.output-summary p{font-size:11px;line-height:1.5;color:#a18b66;margin:8px 0 0}.output-summary p:empty{display:none}.download{width:100%;display:flex;align-items:center;justify-content:center;gap:11px;background:var(--green);color:white;border:0;border-radius:9px;padding:14px;font-size:11px;font-weight:600;margin-top:18px;transition:.15s}.download:hover:not(:disabled){background:#38584b}.download svg{width:16px;height:16px}.download-all{width:100%;border:1px solid #d5dfce;background:#f6f9f1;color:#67815f;border-radius:8px;padding:11px;font-size:12px;margin-top:9px}.download-note{font-size:12px;color:#b0b8aa;margin:9px 0 0;text-align:center}.explain{margin:26px 0;border:1px solid #e3e7dd;border-radius:12px;background:#f0f2e9;padding:18px 22px}.explain-title{display:flex;align-items:center;gap:9px}.explain h3{font-size:11px;font-weight:600;color:#73856c}.info-icon{border:1px solid #abb9a0;color:#97a78a;width:15px;height:15px;border-radius:50%;font:10px Georgia,serif;display:grid;place-items:center}.explain p{font-size:12px;line-height:1.8;color:#97a18e;margin:8px 0 0 24px}.preview-stage{height:305px;border-radius:11px;display:grid;place-items:center;background-color:#f1f3ed;background-image:linear-gradient(45deg,#e7ebe1 25%,transparent 25%),linear-gradient(-45deg,#e7ebe1 25%,transparent 25%),linear-gradient(45deg,transparent 75%,#e7ebe1 75%),linear-gradient(-45deg,transparent 75%,#e7ebe1 75%);background-size:20px 20px;background-position:0 0,0 10px,10px -10px,-10px 0;overflow:hidden;padding:18px}.preview-stage img{max-width:100%;max-height:100%;object-fit:contain;box-shadow:0 3px 15px #304f3520}.image-name-row{display:flex;justify-content:space-between;gap:12px;align-items:center;padding:16px 0}.image-name-row strong{display:block;font-size:11px;font-weight:500;max-width:300px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap}.image-name-row span{font-size:11px;color:#a2ad9d;display:block;margin-top:5px}.remove{border:0;background:#f3f5ed;color:#9ca98f;width:27px;height:27px;border-radius:6px;font-size:20px}.image-queue{display:flex;gap:8px;overflow:auto;padding-bottom:12px}.queue-item{flex-shrink:0;border:1px solid #e6ebdf;border-radius:7px;background:#f8faf4;padding:5px;cursor:pointer;width:56px;height:56px}.queue-item img{width:100%;height:100%;object-fit:contain}.queue-item.active{border:2px solid #78966b;padding:4px}.source-metrics{display:grid;grid-template-columns:1.3fr 1fr 1fr;border-top:1px solid var(--line);padding-top:17px;gap:10px}.source-metrics span{font-size:12px;color:#a6b09e;display:block}.source-metrics strong{font-size:12px;font-weight:500;display:block;margin-top:7px;color:#7d8c72}.print-size-fields{display:grid;grid-template-columns:1fr 1fr;gap:10px;background:#f8faf4;padding:12px;border-radius:9px}.print-size-fields label{font-size:11px;color:#829376}.print-size-fields input,.print-size-fields select{margin-top:6px;padding:9px}.print-size-fields p{grid-column:1/-1;font-size:12px;color:#a3af9c;margin:0;line-height:1.6}.error{padding:10px;border-radius:7px;background:#fcf0e8;color:#a7784e;font-size:12px;line-height:1.7}footer{border-top:1px solid #e3e7dc;padding:19px 0 25px;display:flex;align-items:center;justify-content:space-between;font-size:12px;color:#a9b2a1}.footer-dot{margin:0 9px;color:#c0c7b8}#toast{position:fixed;bottom:24px;left:50%;transform:translate(-50%,12px);background:#334e40;color:white;padding:12px 20px;border-radius:10px;font-size:11px;opacity:0;pointer-events:none;transition:.2s;max-width:90vw;z-index:20}#toast.show{opacity:1;transform:translate(-50%,0)}@media(max-width:700px){header{padding:19px 22px}.private{display:none}.header-links{gap:0}main{padding:0 18px}.intro{padding:32px 0 27px}h1{font-size:35px}.workspace{grid-template-columns:1fr;gap:15px}.image-panel,.settings-panel{padding:22px}.dropzone{min-height:235px}.upload-art{width:69px;height:69px;margin-bottom:15px}.upload-art svg{width:54px;height:54px}.dropzone>p{margin-bottom:16px}.preview-stage{height:265px}.explain{padding:16px}.explain p{margin-left:0}footer>span:last-child{display:none}.image-name-row strong{max-width:240px}}

/* Keep small instructional text readable on light surfaces. */
```

Appendix A - Source code (continued) - lines 4 to 18 of 139

```
4  .private,.intro>p:last-child,.print-badge,.dpi-display span,.dpi-display p,.range-labels,.preset small,.field-help,.output-summary span,.output-summary strong,.download-note,.file-hint,.privacy-note p,.image-name-row span,.source-metrics span,.source-metrics strong,.print-size-fields label,.print-size-fields p,.explain p,footer{color:#647365}.explain h3,.field-label{color:#526849}.dropzone>p{color:#6f7e6e}
5  </style></head><body>
6  <header><a class="brand" href="./"><span class="mark"><svg viewBox="0 0 24 24" fill="none" aria-hidden="true"><path d="M5 3h10l4 4v14H5V3Z" stroke="currentColor" stroke-width="1.6"/><path d="M15 3v5h4M8 12h8M8 16h5" stroke="currentColor" stroke-width="1.6"/></svg></span><span>DPI<span class="brand-light">studio</span></span></a><div class="header-links"><span class="private"><i></i> Images stay on your device</span><a href="DPI-Studio-Offline.html" download class="offline-link">Use offline ↗</a></div></header>
7  <main><div class="intro"><p class="eyebrow">A LITTLE DETAIL. A BETTER PRINT.</p><h1>Make your images<br><span>print-ready.</span></h1><p>Choose your image. Set 300–600 DPI. Download a fresh copy.</p></div>
8  <div class="workspace"><section class="image-panel"><div class="panel-top"><div><span class="step">01</span><h2>Your images</h2></div><button id="add-more" class="quiet" hidden>+ Add more</button></div>
9  <input id="file-input" type="file" accept="image/png,image/jpeg,image/webp" multiple hidden>
10 <button class="dropzone" id="dropzone"><div class="upload-art"><svg viewBox="0 0 80 80" fill="none" aria-hidden="true"><rect x="18" y="12" width="44" height="56" rx="6" stroke="currentColor" stroke-width="1.5"/><path d="M29 49 39 39l8 8 7-7M29 57h25" stroke="currentColor" stroke-width="1.5" stroke-linecap="round"/><circle cx="47" cy="29" r="4" stroke="currentColor" stroke-width="1.5"/><path d="M8 33h-5M77 47h-5M10 68l3-3M67 12l3-3" stroke="currentColor" stroke-width="1.5" stroke-linecap="round"/></svg></div><h3>Drop your images here</h3><p>or <strong>browse your files</strong></p><span class="file-hint">JPG, PNG & WebP · Up to 25 MB each · 10 images</span></button>
11 <div id="preview-panel" hidden><div class="preview-stage" id="preview-stage"><img id="preview-image" alt="Selected image preview"></div><div class="image-name-row"><div><strong id="image-name"></strong><span id="image-details"></span></div><button id="remove-image" class="remove" aria-label="Remove selected image">×</button></div><div id="image-queue" class="image-queue"></div><div class="source-metrics"><div><span>Original size</span><strong id="original-size">—</strong></div><div><span>Detected DPI</span><strong id="original-dpi">—</strong></div><div><span>File size</span><strong id="original-filesize">—</strong></div></div></div>
12 <div class="privacy-note"><svg viewBox="0 0 20 20" fill="none" aria-hidden="true"><path d="M5 8V6a5 5 0 0 1 10 0v2M3 8h14v10H3V8Z" stroke="currentColor" stroke-width="1.4"/></svg><p>Private by design. Images are processed in your browser and are never uploaded.</p></div></section>
13 <section class="settings-panel"><div class="panel-top"><div><span class="step">02</span><h2>Choose your DPI</h2></div><span class="print-badge">PRINT SETTINGS</span></div>
14 <div class="dpi-display"><div><input id="dpi-number" type="number" min="300" max="600" step="1" value="300" aria-label="Output DPI"><span>DPI</span></div><p id="dpi-description">A good starting point for everyday prints.</p></div><input id="dpi-range" type="range" min="300" max="600" step="1" value="300" aria-label="Select output DPI"><div class="range-labels"><span>300 DPI</span><span>600 DPI</span></div><div class="presets"><button class="preset active" data-dpi="300">300 <small>Standard</small></button><button class="preset" data-dpi="450">450 <small>Detailed</small></button><button class="preset" data-dpi="600">600 <small>High density</small></button></div>
15 <div class="field-section"><label class="field-label" for="resize-mode">Pixel dimensions</label><select id="resize-mode"><option value="keep">Keep original pixel dimensions</option><option value="resize">Resize for a chosen print width</option></select><p class="field-help" id="mode-help">Set the print density without resizing the image.</p></div>
16 <div id="print-size-fields" class="print-size-fields" hidden><label for="print-width">Print width<input id="print-width" type="number" min="0.1" max="100" step="0.1" value="6"></label><label for="print-unit">Unit<select id="print-unit"><option value="in">Inches</option><option value="cm">Centimeters</option></select></label><p>Height adjusts automatically to keep the image proportions.</p></div>
17 <div class="field-section"><label class="field-label" for="output-format">Download format</label><select id="output-format"><option value="png">PNG · lossless, preserves transparency</option><option value="jpeg">JPG · smaller file, white background</option></select><p class="field-help" id="format-help">A lossless copy of the decoded image. Original EXIF metadata is not carried over.</p></div>
18 <div class="output-summary"><div><span>Output dimensions</span><strong id="output-pixels">Add an image to see</strong></div><div><span>Print size at chosen DPI</span><strong id="output-print">—</strong></div><p id="scale-note"></p></div><p id="error" class="error" role="alert" hidden></p>
```

Appendix A - Source code (continued) - lines 19 to 40 of 139

```
<button id="download" class="download" disabled><span>Download image</span><svg viewBox="0 0 20 20" fi
ll="none" aria-hidden="true"><path d="M10 2v11m-4-4 4 4 4-4M3 14v4h14v-4" stroke="currentColor" stroke
-width="1.5" stroke-linecap="round" stroke-linejoin="round"/></svg></button><button id="download-all"
class="download-all" hidden>Download all as ZIP</button><p class="download-note">Your original files s
tay untouched.</p></section></div>
<section class="explain"><div class="explain-title"><span class="info-icon">i</span><h3>What does chan
ging DPI do?</h3></div><p>DPI tells printing software how densely to place an image’s pixels. A higher
DPI makes the same pixels print at a smaller size. To keep a larger print size, choose the resize opti
on. Resizing interpolates pixels; it does not recover missing detail or fix blur.</p></section>
<footer><span>DPI Studio <span class="footer-dot">·</span> Made for a clearer workflow.</span><span>No
installation. No image uploads.</span></footer></main><div id="toast" role="status" aria-live="polit
e"></div><script>(function(root){
  'use strict';
  const encoder=new TextEncoder();
  const table=new Uint32Array(256);
  for(let i=0;i<256;i++){let c=i;for(let j=0;j<8;j++)c=(c&1)?0xedb88320^(c>>>1):c>>>1;table[i]=c>>>0;}
  function crc32(bytes){let c=0xffffffff;for(const b of bytes)c=table[(c^b)&255]^(c>>>8);return (c^0xf
fffffff)>>>0;}
  function concat(parts){const out=new Uint8Array(parts.reduce((n,p)=>n+p.length,0));let i=0;for(const
p of parts){out.set(p,i);i+=p.length;}return out;}
  function validDpi(dpi){if(!Number.isInteger(dpi)||dpi<300||dpi>600)throw Error('Choose a whole-numbe
r DPI between 300 and 600.');return dpi;}
  function isPng(b){return b.length>=33&&[137,80,78,71,13,10,26,10].every((x,i)=>b[i]===x);}
  function pngChunks(b){if(!isPng(b))throw Error('Not a valid PNG image.');const view=new DataView(b.b
uffer,b.byteOffset,b.byteLength);const chunks=[];let i=8;while(i+12<=b.length){const len=view.getUint3
2(i);if(i+len+12>b.length)throw Error('The PNG file is incomplete.');const type=String.fromCharCode
(...b.subarray(i+4,i+8));chunks.push({start:i,length:len,type,end:i+len+12});i+=len+12;if(type==='IEN
D')return chunks;}throw Error('The PNG file has no end marker.');}
  function pngDpi(bytes,dpi){validDpi(dpi);const b=new Uint8Array(bytes),chunks=pngChunks(b);const dat
a=new Uint8Array(9),v=new DataView(data.buffer);const ppm=Math.round(dpi/0.0254);v.setUint32(0,ppm);v.
setUint32(4,ppm);data[8]=1;
    const body=concat([encoder.encode('pHYs'),data]),chunk=new Uint8Array(21),cv=new DataView(chunk.bu
ffer);cv.setUint32(0,9);chunk.set(body,4);cv.setUint32(17,crc32(body));
    const parts=[b.subarray(0,8)];let added=false;
    for(const c of chunks){if(c.type==='pHYs')continue;parts.push(b.subarray(c.start,c.end));if(c.type
===='IHDR'){parts.push(chunk);added=true;}}
    if(!added)throw Error('The PNG image header is missing.');return concat(parts);
  }
  function jpegSegments(b){if(b.length<4||b[0]!==255||b[1]!==216)throw Error('Not a valid JPEG imag
e.');const v=new DataView(b.buffer,b.byteOffset,b.byteLength),parts=[];let i=2;while(i<b.length){const
start=i;if(b[i]!==255)throw Error('Invalid JPEG header.');while(i<b.length&&b[i]===255)i++;const marke
r=b[i++];if(marker===0xda||marker===0xd9){parts.push({start,marker,end:b.length,scan:true});return par
ts;}if(marker===0x01||(marker>=0xd0&&marker<=0xd7)){parts.push({start,marker,end:i});continue;}if(i+2>
b.length)throw Error('Incomplete JPEG segment.');const len=v.getUint16(i);if(len<2||i+len>b.length)thr
ow Error('Invalid JPEG segment length.');parts.push({start,marker,data:i+2,end:i+len,length:len});i+=l
en;}throw Error('The JPEG image is incomplete.');}
  function jpegDpi(bytes,dpi){validDpi(dpi);const b=new Uint8Array(bytes),segments=jpegSegments(b),par
ts=[b.subarray(0,2)];let found=false;for(const s of segments){let p=b.slice(s.start,s.end);if(s.marker
===0xe0&&s.length>=16&&String.fromCharCode(...b.subarray(s.data,s.data+5))==='JFIF\0'){const j=s.data-
s.start;const v=new DataView(p.buffer);p[j+7]=1;v.setUint16(j+8,dpi);v.setUint16(j+10,dpi);found=tru
e;}parts.push(p);}if(!found){const app0=new Uint8Array([255,224,0,16,74,70,73,70,0,1,2,1,0,0,0,0,0,
0]);const v=new DataView(app0.buffer);v.setUint16(12,dpi);v.setUint16(14,dpi);parts.splice(1,0,app0);}
return concat(parts);}
  function readDpi(bytes){const b=new Uint8Array(bytes);try{if(isPng(b)){const v=new DataView(b.buffe
r,b.byteOffset,b.byteLength);for(const c of pngChunks(b)){if(c.type==='pHYs'&&c.length===9&&b[c.start+
16]===1)return v.getUint32(c.start+8)*.0254;}return null;}
    if(b[0]===255&&b[1]===216){const v=new DataView(b.buffer,b.byteOffset,b.byteLength);for(const s of
jpegSegments(b)){if(s.marker===0xe0&&s.length>=16&&String.fromCharCode(...b.subarray(s.data,s.data+5))
==='JFIF\0'){const units=b[s.data+7],n=v.getUint16(s.data+8);if(units===1&&n)return n;if(units===2&&n)
return n*2.54;}}}}
```

```
    catch{}return null;
  }
  function dimensions(width,height,dpi,mode='keep',printWidth=6,unit='in'){
    validDpi(dpi);if(!(width>0&&height>0))throw Error('Image dimensions are invalid.');
    let w=width,h=height;if(mode==='resize'){const inches=unit==='cm'?Number(printWidth)/2.54:Number(printWidth);if(!Number.isFinite(inches)||inches<=0)throw Error('Enter a positive print width.');w=Math.max(1,Math.round(inches*dpi));h=Math.max(1,Math.round(w*height/width));}
    if(w>16384||h>16384||w*h>40000000)throw Error('This size is too large for a browser export. Use a smaller print width or DPI.');
    return {width:w,height:h,printWidth:w/dpi,printHeight:h/dpi,scale:w/width};
  }
  function zip(files){
    if(files.length>100)throw Error('Too many ZIP entries.');const entries=[],central=[];let offset=0;const now=new Date(),dosTime=(now.getHours()<<11)|(now.getMinutes()<<5)|(now.getSeconds()>>1),dosDate=((Math.max(1980,now.getFullYear())-1980)<<9)|((now.getMonth()+1)<<5)|now.getDate();
    for(const file of files){const name=encoder.encode(file.name),data=new Uint8Array(file.data),crc=crc32(data),local=new Uint8Array(30+name.length),v=new DataView(local.buffer);v.setUint32(0,0x04034b50,true);v.setUint16(4,20,true);v.setUint16(6,0x0800,true);v.setUint16(10,dosTime,true);v.setUint16(12,dosDate,true);v.setUint32(14,crc,true);v.setUint32(18,data.length,true);v.setUint32(22,data.length,true);v.setUint16(26,name.length,true);local.set(name,30);entries.push(local,data);
      const entry=new Uint8Array(46+name.length),e=new DataView(entry.buffer);e.setUint32(0,0x02014b50,true);e.setUint16(4,20,true);e.setUint16(6,20,true);e.setUint16(8,0x0800,true);e.setUint16(12,dosTime,true);e.setUint16(14,dosDate,true);e.setUint32(16,crc,true);e.setUint32(20,data.length,true);e.setUint32(24,data.length,true);e.setUint16(28,name.length,true);e.setUint32(42,offset,true);entry.set(name,46);central.push(entry);offset+=local.length+data.length;
    }
    const centralBytes=concat(central),end=new Uint8Array(22),v=new DataView(end.buffer);v.setUint32(0,0x06054b50,true);v.setUint16(8,files.length,true);v.setUint16(10,files.length,true);v.setUint32(12,centralBytes.length,true);v.setUint32(16,offset,true);return concat([...entries,centralBytes,end]);
  }
  const api={crc32,pngDpi,jpegDpi,readDpi,dimensions,zip,pngChunks,validDpi};root.DpiCore=api;if(typeof module!=='undefined'&&module.exports)module.exports=api;
})(typeof globalThis!=='undefined'?globalThis:this);
</script><script>'use strict';
const $=s=>document.querySelector(s),$$=s=>[...document.querySelectorAll(s)];
const files=[];let selected=-1,busy=false,toastTimer;
const state={dpi:300,mode:'keep',printWidth:6,unit:'in',format:'png'};
const esc=x=>String(x).replace(/[&<>"']/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;',"'":'&#39;'}[c]));
function toast(text){$('#toast').textContent=text;$('#toast').classList.add('show');clearTimeout(toastTimer);toastTimer=setTimeout(()=>$('#toast').classList.remove('show'),4000);}
function error(text){$('#error').textContent=text;$('#error').hidden=!text;}
function sizeLabel(n){return n>=1048576?(n/1048576).toFixed(1)+' MB':(n/1024).toFixed(1)+' KB';}
function image(url){return new Promise((resolve,reject)=>{const i=new Image();i.onload=()=>resolve(i);i.onerror=()=>reject(Error('This image could not be opened. Try a standard JPG, PNG, or WebP file.'));i.src=url;});}
function plan(file=files[selected]){if(!file)return null;return DpiCore.dimensions(file.width,file.height,state.dpi,state.mode,state.printWidth,state.unit);}
function controlsDisabled(value){busy=value;$('#download').disabled=value||selected<0;$('#download-all').disabled=value;$('#file-input').disabled=value;$('#add-more').disabled=value;$('#remove-image').disabled=value;$$('.preset,input,select,.queue-item').forEach(x=>x.disabled=value);}
function render(){
  const file=files[selected];$('#dropzone').hidden=!!file;$('#preview-panel').hidden=!file;$('#add-more').hidden=!file;$('#download-all').hidden=files.length<2;
  if(file){$('#preview-image').src=file.url;$('#image-name').textContent=file.file.name;$('#image-details').textContent=`Image ${selected+1} of ${files.length} · ${file.file.type.split('/')[1].toUpperCase()}`;$('#original-size').textContent=`${file.width.toLocaleString()} × ${file.height.toLocaleString()} px`;$('#original-dpi').textContent=file.dpi?Math.round(file.dpi)+' DPI':'Not detected';$('#original-filesize').textContent=sizeLabel(file.file.size);}
  else{$('#preview-image').removeAttribute('src');}
```

Appendix A - Source code (continued) - lines 73 to 109 of 139

```
  $('#image-queue').innerHTML=files.length>1?files.map((f,i)=>`<button class="queue-item ${i===selected?'active':''}" data-index="${i}" title="${esc(f.file.name)}" aria-label="Select ${esc(f.file.name)}" ${busy?'disabled':''}><img src="${esc(f.thumb)}" alt=""></button>`).join(''):'';
  $('#download-all').textContent=`Download all ${files.length} images as ZIP`;
  update();
}
function update(){
  $('#dpi-range').value=state.dpi;$('#dpi-number').value=state.dpi;$('#dpi-range').style.background=`linear-gradient(to right,#486b5b ${(state.dpi-300)/3}%,#e6eadd ${(state.dpi-300)/3}%)`;
  $$('.preset').forEach(p=>p.classList.toggle('active',Number(p.dataset.dpi)===state.dpi));
  $('#dpi-description').textContent=state.dpi===300?'A good starting point for everyday prints.':state.dpi===600?'More pixels per printed inch. Check the print size below.':'Choose the density your printer or publisher requires.';
  $('#print-size-fields').hidden=state.mode!=='resize';$('#mode-help').textContent=state.mode==='keep'?'Set the print density without resizing the image.':'Calculate new pixel dimensions from the print width and DPI.';
  $('#format-help').textContent=state.format==='png'?'A lossless copy of the decoded image. Original EXIF metadata is not carried over.':'High-quality JPEG export. Transparent areas become white; original EXIF metadata is not carried over.';
  try{
    const p=plan();$('#output-pixels').textContent=p?`${p.width.toLocaleString()} × ${p.height.toLocaleString()} px`:'Add an image to see';
    $('#output-print').textContent=p?`${p.printWidth.toFixed(2)} × ${p.printHeight.toFixed(2)} in · ${(p.printWidth*2.54).toFixed(1)} × ${(p.printHeight*2.54).toFixed(1)} cm`:'—';
    $('#scale-note').textContent=p&&state.mode==='resize'?(p.scale>1.01?`Enlargement: ${p.scale.toFixed(2)}×. New pixels are interpolated; missing detail cannot be restored.`:p.scale<.99?'The image will be reduced to match this print size.':'No meaningful enlargement is needed.') :'';
    $('#download').disabled=busy||!p;$('#download-all').disabled=busy||!p;error('');
  }catch(e){$('#output-pixels').textContent='Invalid output size';$('#output-print').textContent='—';$('#download').disabled=true;$('#download-all').disabled=true;error(e.message);}
}
async function addFiles(list){
  if(busy)return;controlsDisabled(true);let added=0;
  try{
    for(const file of Array.from(list)){
      if(files.length>=10){toast('You can work with up to 10 images at a time.');break;}
      if(!['image/png','image/jpeg','image/webp'].includes(file.type)){toast(`${file.name}: use JPG, PNG, or WebP.`);continue;}
      if(file.size>25*1048576){toast(`${file.name} is larger than 25 MB.`);continue;}
      if(files.reduce((n,x)=>n+x.file.size,0)+file.size>100*1048576){toast('This batch exceeds 100 MB. Remove an image before adding more.');break;}
      const url=URL.createObjectURL(file);
      try{
        const img=await image(url);if(img.naturalWidth*img.naturalHeight>40000000)throw Error(`${file.name}: images must be 40 megapixels or smaller.`);
        const raw=new Uint8Array(await file.arrayBuffer());
        if(file.type==='image/png'&&DpiCore.pngChunks(raw).some(c=>c.type==='acTL'))throw Error('Animated PNG is not supported. Choose a still image.');
        if(file.type==='image/webp'&&raw.length>21&&String.fromCharCode(...raw.subarray(12,16))==='VP8X'&&(raw[20]&2))throw Error('Animated WebP is not supported. Choose a still image.');
        const canvas=document.createElement('canvas'),ratio=Math.min(1,96/Math.max(img.naturalWidth,img.naturalHeight));canvas.width=Math.max(1,Math.round(img.naturalWidth*ratio));canvas.height=Math.max(1,Math.round(img.naturalHeight*ratio));canvas.getContext('2d').drawImage(img,0,0,canvas.width,canvas.height);
        files.push({file,url,width:img.naturalWidth,height:img.naturalHeight,dpi:DpiCore.readDpi(raw),thumb:canvas.toDataURL('image/png')});img.src='';selected=files.length-1;added++;
      }catch(e){URL.revokeObjectURL(url);toast(e.message);}
    }
  }finally{controlsDisabled(false);$('#file-input').value='';render();if(added)toast(`${added} image${added>1?'s':''} ready. Choose 300–600 DPI.`);}
}
```

Appendix A - Source code (continued) - lines 110 to 137 of 139

```
function toBlob(canvas,type){return new Promise((resolve,reject)=>canvas.toBlob(blob=>blob?resolve(blob):reject(Error('The browser could not export this image. Try a smaller size.')),type,.98));}
async function exportFile(file,index=selected){
  const p=plan(file),img=await image(file.url),canvas=document.createElement('canvas');canvas.width=p.width;canvas.height=p.height;const ctx=canvas.getContext('2d');if(!ctx)throw Error('Image export is unavailable in this browser.');
  if(state.format==='jpeg'){ctx.fillStyle='#ffffff';ctx.fillRect(0,0,p.width,p.height);}
  ctx.imageSmoothingEnabled=true;ctx.imageSmoothingQuality='high';ctx.drawImage(img,0,0,p.width,p.height);img.src='';
  const blob=await toBlob(canvas,state.format==='png'?'image/png':'image/jpeg');const bytes=new Uint8Array(await blob.arrayBuffer());const data=state.format==='png'?DpiCore.pngDpi(bytes,state.dpi):DpiCore.jpegDpi(bytes,state.dpi);
  const actual=DpiCore.readDpi(data);if(!actual||Math.abs(actual-state.dpi)>.02)throw Error('The DPI metadata could not be verified.');
  canvas.width=1;canvas.height=1;
  const base=file.file.name.replace(/\.[^.]+$/,'').replace(/[<>:"/\\|?*\x00-\x1f]/g,'_').slice(0,100)||'image';
  return {name:`${base}_${state.dpi}dpi${files.length>1?'_'+(index+1):''}.${state.format==='png'?'png':'jpg'}`,data,type:state.format==='png'?'image/png':'image/jpeg'};
}
function download(data,name,type){const url=URL.createObjectURL(new Blob([data],{type})),a=document.createElement('a');a.href=url;a.download=name;a.click();setTimeout(()=>URL.revokeObjectURL(url),60000);}
async function runExport(all=false){
  if(busy||selected<0)return;controlsDisabled(true);error('');const button=all?$('#download-all'):$('#download'),old=button.innerHTML;let failure=null;
  try{
    if(all){const outputs=[];let bytes=0;for(let i=0;i<files.length;i++){button.textContent=`Preparing image ${i+1} of ${files.length}…`;const out=await exportFile(files[i],i);bytes+=out.data.length;if(bytes>200*1048576)throw Error('This ZIP would exceed 200 MB. Download the images individually or use smaller dimensions.');outputs.push(out);}download(DpiCore.zip(outputs),`DPI-Studio_${state.dpi}dpi.zip`,'application/zip');toast(`All ${files.length} images exported at ${state.dpi} DPI.`);}
    else{button.textContent='Preparing your image…';const out=await exportFile(files[selected]);download(out.data,out.name,out.type);toast(`Downloaded at ${state.dpi} DPI. Original image untouched.`);}
  }catch(e){failure=e.message;}finally{button.innerHTML=old;controlsDisabled(false);update();if(failure)error(failure);}
}
$('#dropzone').onclick=()=>$('#file-input').click();$('#add-more').onclick=()=>$('#file-input').click();$('#file-input').onchange=e=>void addFiles(e.target.files);
const drop=$('#dropzone');for(const type of ['dragenter','dragover'])drop.addEventListener(type,e=>{e.preventDefault();drop.classList.add('dragging');});for(const type of ['dragleave','drop'])drop.addEventListener(type,e=>{e.preventDefault();drop.classList.remove('dragging');});drop.addEventListener('drop',e=>void addFiles(e.dataTransfer.files));
document.addEventListener('dragover',e=>e.preventDefault());document.addEventListener('drop',e=>{e.preventDefault();if(!drop.contains(e.target))void addFiles(e.dataTransfer.files);});
$('#remove-image').onclick=()=>{if(busy||selected<0)return;URL.revokeObjectURL(files[selected].url);files.splice(selected,1);selected=Math.min(selected,files.length-1);render();};
$('#image-queue').onclick=e=>{const b=e.target.closest('[data-index]');if(b&&!busy){selected=Number(b.dataset.index);render();}};
function setDpi(value){if(busy)return;const n=Number(value);if(!Number.isInteger(n)||n<300||n>600){error('Enter a whole-number DPI from 300 to 600.');$('#download').disabled=true;$('#download-all').disabled=true;return;}state.dpi=n;update();}
$('#dpi-range').oninput=e=>setDpi(e.target.value);$('#dpi-number').onchange=e=>setDpi(e.target.value);$$('.preset').forEach(b=>b.onclick=()=>setDpi(b.dataset.dpi));
$('#resize-mode').onchange=e=>{state.mode=e.target.value;update();};$('#print-width').oninput=e=>{state.printWidth=Number(e.target.value);update();};$('#print-unit').onchange=e=>{const old=state.unit;state.unit=e.target.value;if(old!==state.unit){state.printWidth=state.unit==='cm'?state.printWidth*2.54:state.printWidth/2.54;$('#print-width').value=state.printWidth.toFixed(2);}update();};$('#output-format').onchange=e=>{state.format=e.target.value;update();};
$('#download').onclick=()=>void runExport(false);$('#download-all').onclick=()=>void runExport(true);
```

Appendix A - Source code (continued) - lines 138 to 139 of 139

```
138 window.addEventListener('beforeunload',()=>files.forEach(f=>URL.revokeObjectURL(f.url)));if(location.p
    rotocol==='file:')$('.offline-link').hidden=true;render();
139 </script></body></html>
```

APPENDIX B

Glossary

Term	Meaning, in plain language
DPI (dots per inch)	The print-density instruction stored in an image file: how many pixels the printer should place in each inch of paper. Higher DPI = the same pixels printed smaller and denser.
PPI (pixels per inch)	The same measurement described from the pixel side, used for screens and scanners. For the files DPI Studio handles, DPI and PPI refer to the same stored value; print has simply kept the older word "dots".
pHYs chunk	The part of a PNG file storing physical pixel dimensions: pixels-per-unit horizontally and vertically, plus a unit flag. DPI Studio writes pixels per metre (DPI / 0.0254), flag set to "metre", protected by a CRC32 checksum.
JFIF / APP0	The JPEG header segment (JFIF - JPEG File Interchange Format) carrying the density: a units byte (1 = dots per inch) and X/Y density values. DPI Studio edits these fields, or inserts a complete APP0 segment when a file has none.
Interpolation	Estimating new pixels from existing neighbours when an image is resized. Smoothing makes the estimate as smooth as possible, but interpolated pixels are estimates - they cannot recreate detail the original never captured.
CRC32	A checksum algorithm used to prove data has been written correctly. It appears twice in DPI Studio: inside PNG chunks, and for every entry in the ZIP writer.
EXIF	Extra data many cameras embed in a file - camera model, settings, time, sometimes location. DPI Studio's output does not carry EXIF over (see Limitations); the original file keeps it.
Lossless / lossy	PNG saving is lossless: decoding the file returns exactly the pixels that were encoded. JPEG saving is lossy: it discards a little information to shrink the file - DPI Studio uses a high quality setting (0.98) to keep that loss minimal.

Troubleshooting - questions and answers

The printed image came out smaller than expected. That is the DPI relationship working as designed: at a higher DPI the same pixels occupy fewer inches. If a specific print size is required, use "Resize for a chosen print width" so the pixel count is recalculated for that size at the chosen DPI.

My publisher still says the file is not 300 DPI. First confirm the downloaded copy - not the original - was sent; originals are never modified. Then check the value shown when the downloaded file is loaded back into DPI Studio: its "Detected DPI" should read 300. If the publisher measures a TIFF workflow, note that DPI Studio outputs PNG and JPG only.

"The DPI metadata could not be verified." The export's readback check failed, so no file was downloaded - the safety net working as intended. Try the other output format (PNG instead of JPG or vice versa); if it persists, the source image may be in an unusual variant - re-save it in any standard editor and try again.

Troubleshooting - more answers

An image will not load. The usual causes are named in the error or toast message: the file is over 25 MB, over 40 megapixels, an animated PNG or WebP, or not a JPG/PNG/WebP at all. Animated images must be replaced with a still frame; oversize images should be reduced in an editor first.

"This size is too large for a browser export." The requested combination - print width times DPI - exceeds 16,384 pixels on a side or 40 megapixels in total. Lower the DPI, narrow the print width, or split the job: this ceiling is the browser's practical memory limit, enforced early so the tab cannot run out of memory mid-export.

The ZIP download was refused. The batch would exceed the 200 MB ZIP cap. Download the images individually, or reduce the resize settings and prepare the ZIP again.

My photo's camera details (date, location) disappeared. Expected behaviour: output files do not carry EXIF metadata (Section 8). The original photo still holds all of it - DPI Studio never touches originals.

Transparent areas turned white in my JPG. JPEG cannot store transparency, so transparent pixels are placed on white - the interface notes this before you choose the format. Choose PNG output when transparency must be preserved.

Can DPI Studio make my small image suitable for a large poster? It can produce a file with the poster's dimensions and density, by interpolation - and its summary will state the enlargement factor plainly. Whether the result looks acceptable at viewing distance is a judgement only the user can make by inspecting the enlarged copy; the tool will not pretend the missing detail has been recovered.

Does it work without internet? Yes - completely. Copy the file to the device once, and every feature works with the connection off, on every system listed in Section 11.

> **End of document.** DPI Studio - designed and generated by **Dr. Asif Ahmad**. Documentation, workflow, technical method and complete source code, as shipped. The tool writes correct print-density metadata and resizes by interpolation; it does not enhance image quality - and it is documented here, in full, so that anyone can verify exactly that.